

1. Project Summary and Goal

The objective of this project was to implement a **Long Short-Term Memory (LSTM)** neural network to perform binary sentiment classification on the **IMDb Movie Reviews dataset**. The model was built using word embeddings trained from scratch and rigorously evaluated against a simple machine learning **baseline** to validate the deep learning approach.

2. Data Preparation and Visualization

The dataset used was the standard IMDb set, which is perfectly balanced.

- Statistical Analysis:** A statistical analysis of all review lengths was performed, finding that the 95th percentile occurred at **598 tokens**. This value was chosen as the Maximum Sequence Length (MAX_LEN) for padding and truncation.
- Word Embeddings Visualization:** A **PCA (Principal Component Analysis)** was successfully executed on the top 500 learned word vectors. The visualization successfully reduced the 128-dimensional vectors to 2D space, though it shows little immediate clustering of related sentiment terms, suggesting the embeddings did not fully mature during training.

3. Model Performance (LSTM)

The LSTM model was trained for 10 epochs. The results indicated a catastrophic failure to generalize, likely due to severe training instability.

Metric	Score	Finding			
Test Accuracy	0.502	The model performed only slightly better than random chance.			
Test F1-Score	0.0552	The F1-score confirms the model is unreliable, especially for the positive class.			
Validation Loss	Increased from 0.6928 to 0.7728	The rapid divergence of Validation Loss indicates severe overfitting right after Epoch 3.			

The **Confusion Matrix** reveals the specific failure mode: the model became heavily biased toward the Negative class (0).

- It correctly identified 97% of the actual Negative reviews (True Negative: 12,186).
- It incorrectly classified 97% of the actual **Positive reviews** as Negative (False Negative: 12,136), effectively ignoring the positive sentiment.

4. Baseline Comparison and Conclusion

The LSTM model's performance was compared against a simpler **Logistic Regression** baseline trained on the same padded data.

Model	Accuracy	F1-Score	

LSTM	0.502	0.0552	
Logistic Regression	0.5087	0.4593	

Conclusion: The Logistic Regression baseline **significantly outperformed** the complex LSTM model.

The most critical observation for documentation is that the LSTM failed to learn due to **overfitting** and **training instability**. This result strongly suggests that the high MAX_LEN of 598, combined with training embeddings from scratch on a complex recurrent network, was too computationally demanding and unstable for the defined hyperparameters.